

Machine Learning for Electrocatalyst Discovery in Energy Conversion: A Survey

Electrocatalyst discovery is central to energy conversion technologies such as water splitting, fuel cells, metal-air batteries, electrochemical carbon dioxide reduction, nitrogen reduction, and electrolyzers. Machine learning can accelerate this search by predicting adsorption energies, overpotentials, selectivity, stability, active-site motifs, synthesis feasibility, and reaction pathways from DFT, high-throughput databases, surface structures, literature, and experiments. This survey synthesizes an arXiv-derived corpus of 101 papers collected during this run and organizes the literature around descriptors, adsorption energies, high-throughput screening, graph and deep learning, active learning, reaction families, catalyst classes, stability, interfaces, datasets, synthesis, and validation. The survey argues that useful machine learning for electrocatalyst discovery must combine physically meaningful targets, uncertainty-aware search, surface and electrolyte context, and experimental validation under realistic operating conditions.

Keywords:

electrocatalyst discovery; machine learning; energy conversion; adsorption energy; high-throughput screening; active learning; graph neural networks; catalyst design

1 Introduction

Electrocatalysts mediate the conversion between electrical energy and chemical bonds. They determine the efficiency, selectivity, durability, and cost of water splitting, fuel cells, electrolyzers, carbon dioxide reduction, nitrogen reduction, metal-air batteries, and related energy-conversion systems. The discovery problem is difficult because useful catalysts must satisfy many constraints at once: active sites should bind intermediates neither too weakly nor too strongly, surfaces should remain stable under potential and electrolyte, products should be selective, and materials should be abundant, synthesizable, and durable [20, 23, 37, 48, 59, 65, 75, 95].

Machine learning is attractive because the electrocatalyst design space is combinatorial. Candidate materials include single-atom catalysts, alloys, high-entropy alloys, oxides, sulfides, nitrides, carbides, MXenes, metal-organic frameworks, perovskites, two-dimensional materials, and defect-engineered surfaces. Each candidate can expose multiple facets, terminations, oxidation states, dopants,

strain states, coverages, and solvent environments. Exhaustive first-principles and experimental screening is therefore impractical [23, 37, 40, 53, 59, 65, 75, 91].

At the same time, electrocatalysis is not a generic materials-prediction task. Activity depends on reaction networks, adsorbate coverage, applied potential, pH, electrolyte, solvation, mass transport, and surface reconstruction. A model that predicts a clean-surface adsorption energy may accelerate screening, but it does not by itself prove catalytic performance. This survey therefore emphasizes the full discovery workflow: descriptors, high-throughput DFT, graph learning, active learning, reaction-specific targets, catalyst classes, stability, synthesis, datasets, and validation.

2 Background

2.1 Electrocatalytic Energy Conversion

Energy-conversion electrocatalysis includes hydrogen evolution and oxygen evolution for water splitting, oxygen reduction for fuel cells and metal-air batteries, carbon dioxide reduction for carbon utilization, nitrogen reduction for ammonia synthesis, and related electrochemical transformations. These reactions are governed by adsorption free energies, electron-proton transfer, surface coverage, solvent and electrolyte effects, and competing pathways. The same catalyst may perform differently under different potential, pH, or electrolyte conditions [13, 19, 20, 23, 40, 68, 75, 91].

2.2 Descriptors and Scaling Relations

Classical electrocatalyst design uses descriptors such as adsorption energies, d-band centers, coordination numbers, charge states, generalized coordination, and reaction free-energy steps. Volcano plots and scaling relations reduce complex reaction networks to searchable trends. Machine learning extends these ideas by learning descriptors from composition, structure, surface graphs, local environments, or text, and by predicting adsorption energies or activity proxies across larger material spaces [13, 23, 40, 53, 68, 75, 91, 98].

2.3 Why Discovery Is Multi-Objective

The best catalyst is rarely the material with the lowest predicted overpotential in a static calculation. Discovery must also account for selectivity, stability, abundance, conductivity, synthesis, poisoning, reconstruction, and integration with electrodes and electrolytes. For CO₂ reduction, selectivity across many carbon products may dominate. For ORR and OER, stability and precious-metal loading matter. For HER, pH and surface state can change mechanisms. Machine learning systems should therefore treat electrocatalyst discovery as constrained, multi-objective decision making.

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined electrocatalyst, electrocatalysis, energy conversion, materials discovery, machine learning, high-throughput DFT, adsorption energy, binding energy,

overpotential, selectivity, volcano relations, HER, OER, ORR, CO₂RR, NRR, water splitting, fuel-cell catalysts, electrolyzer catalysts, single-atom catalysts, high-entropy alloys, MXenes, perovskites, MOFs, two-dimensional materials, graph neural networks, deep learning, foundation models, machine-learning potentials, active learning, Bayesian optimization, autonomous experiments, microkinetics, stability, surface Pourbaix analysis, synthesis, and literature mining. Records were retained when title or abstract evidence explicitly connected electrocatalytic energy conversion with catalyst materials and machine-learning, DFT, screening, optimization, or validation methods. Duplicates were merged by arXiv identifier, and the final bibliography cites 101 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *reaction family*: HER, OER, ORR, CO₂RR, NRR, water splitting, fuel-cell catalysis, electrolyzer catalysis, and coupled electrochemical conversions. The second is *catalyst class*: metals, alloys, high-entropy alloys, single-atom catalysts, oxides, sulfides, nitrides, carbides, MXenes, perovskites, MOFs, and two-dimensional materials. The third is *target*: adsorption energy, free-energy step, overpotential, selectivity, stability, conductivity, synthesis, or measured activity. The fourth is *representation*: composition descriptor, surface descriptor, adsorbate-site graph, crystal graph, local environment, reaction graph, text, or interatomic potential. The fifth is *learning task*: regression, classification, ranking, surrogate modeling, active learning, Bayesian optimization, generative design, or literature mining. The sixth is *data source*: DFT, microkinetics, high-throughput databases, experiments, literature, or closed-loop measurements. The seventh is *validation*: random split, catalyst-family holdout, reaction holdout, prospective DFT, synthesis, electrochemical testing, or durability testing. The eighth is *deployment*: uncertainty, interpretability, reproducibility, synthesis constraints, and integration with experimental workflows.

3.3 Descriptors, Adsorption Energies, and Activity Prediction

Descriptors remain the conceptual bridge between catalysis knowledge and machine learning. Adsorption energies and reaction free energies are common because they are computable and connect to volcano relations. However, many failures arise when a descriptor ignores coverage, solvent, potential, or

surface reconstruction. Learning systems should report which physical assumptions define the target label and whether the label is meant to approximate activity, selectivity, or only one elementary step [13, 23, 40, 53, 68, 75, 91, 98].

3.4 High-Throughput DFT and Materials-Informatics Screening

High-throughput screening enumerates candidate surfaces, computes adsorption energies or stability proxies, and ranks materials. Machine learning can accelerate this pipeline by predicting expensive DFT quantities, guiding which sites to evaluate, correcting systematic errors, or replacing parts of the calculation with a surrogate. The challenge is that screening funnels can amplify errors: a bad early filter may discard the most interesting candidates [13, 19, 23, 40, 53, 68, 75, 91].

3.5 Graph Learning, Deep Models, and Machine-Learning Potentials

Graph neural networks encode atoms, bonds, surface neighborhoods, adsorbates, and reaction environments. They can share statistical strength across catalyst families and support predictions for adsorption, stability, and reaction energetics. Machine-learning potentials can accelerate relaxation, molecular dynamics, and sampling under complex surface conditions. These models should be evaluated on chemically meaningful splits, such as held-out elements, facets, adsorbates, or reaction families [13, 30, 36, 63, 77, 85, 91, 95].

3.6 Active Learning, Bayesian Optimization, and Autonomous Experiments

Active learning and Bayesian optimization are natural when each DFT calculation or experiment is costly. Acquisition functions can balance predicted activity, uncertainty, diversity, stability, and synthesis feasibility. In autonomous experiments, the model must also handle failed syntheses, noisy electrochemical measurements, batch constraints, and changing experimental priorities. Negative results are valuable because they teach the model which regions of catalyst space are unproductive [13, 15, 19, 20, 59, 63, 65, 75].

3.7 Reaction Families

Different reactions demand different targets. HER often centers on hydrogen adsorption and pH-

dependent mechanisms. OER requires multiple oxygenated intermediates and strong stability constraints. ORR involves selectivity between two- and four-electron pathways. CO₂RR has many products and severe selectivity challenges. NRR must distinguish real ammonia production from contamination and competing hydrogen evolution. These differences make reaction-specific validation essential [13, 19, 20, 23, 40, 68, 75, 91].

3.8 Catalyst Materials and Active-Site Classes

Catalyst classes impose different priors. Single-atom catalysts require coordination and support descriptors. Alloys and high-entropy alloys require composition and surface segregation modeling. MXenes and two-dimensional materials expose termination and defect variables. Perovskites and oxides involve oxidation states and lattice oxygen chemistry. MOFs and derived catalysts connect porous structures to active sites. Machine learning should exploit these priors without overfitting to one family [23, 37, 40, 53, 59, 65, 75, 91].

3.9 Stability, Selectivity, Interfaces, and Microkinetics

Activity is only one part of catalyst utility. Stability under potential, dissolution, oxidation, poisoning, reconstruction, electrolyte effects, and interface structure can dominate practical performance. Selectivity depends on competing pathways and coverage. Microkinetic models can connect elementary steps to measurable rates, but they require many parameters. Learning these quantities demands data that encode potential, pH, electrolyte, and operating conditions instead of only clean-surface descriptors [14, 19, 23, 37, 53, 68, 69, 98].

3.10 Datasets, Synthesis, and Validation

Electrocatalyst datasets mix DFT calculations, curated literature, and experiments. Each source has different biases: DFT may use idealized surfaces, literature may overrepresent positive results, and experiments vary in normalization and protocol. Strong validation should include held-out catalyst families, prospective calculations, synthesis, activity tests, selectivity measurements, and durability. Benchmarks should measure discovery utility, not only regression error [13, 19, 20, 23, 53, 65, 91, 98].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across descriptors, screening, deep learning, active learning, reaction families, catalyst classes, stability, datasets, synthesis, and validation.

Descriptors, adsorption energies, and activity prediction The descriptors, adsorption energies, and activity prediction cluster contains work that develops methods, systems, datasets, or analyses central to this survey [28, 40, 51, 55, 63, 83, 88, 91, 98]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [17, 31, 78, 89, 92].

High-throughput dft and materials-informatics screening The high-throughput DFT and materials-informatics screening cluster contains work that develops methods, systems, datasets, or analyses central to this survey [15, 19, 53, 68, 69, 74, 75, 77, 95]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [2, 6, 8, 16, 29, 58, 81, 84, 101].

Graph learning, deep models, and machine-learning potentials The graph learning, deep models, and machine-learning potentials cluster contains work that develops methods, systems, datasets, or analyses central to this survey [18, 30, 36, 50, 56, 72, 76].

Active learning, bayesian optimization, and autonomous experiments The active learning, Bayesian optimization, and autonomous experiments cluster contains work that develops methods, systems, datasets, or analyses central to this survey [33].

Reaction families for energy conversion The reaction families for energy conversion cluster contains work that develops methods, systems, datasets, or analyses central to this survey [10, 13, 23, 26, 37, 48, 59, 65, 90]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [1, 25, 42, 43, 66, 70, 79, 86, 93]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 34, 38, 41, 60, 61, 87, 96, 99]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 5, 9, 35, 44, 45, 47, 54, 94]. Additional retained studies broaden the cluster with alter-

native architectures, training signals, data sources, and deployment assumptions [49].

Catalyst materials and active-site classes The catalyst materials and active-site classes cluster contains work that develops methods, systems, datasets, or analyses central to this survey [7, 67, 97].

Stability, selectivity, interfaces, and microkinetics The stability, selectivity, interfaces, and microkinetics cluster contains work that develops methods, systems, datasets, or analyses central to this survey [21, 24, 32, 39, 57, 62, 71, 73, 100]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [11, 27, 64, 82].

Datasets, synthesis, validation, and experimental deployment The datasets, synthesis, validation, and experimental deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [4, 14, 20, 22, 46, 52, 80, 85].

4 Challenges and Prospects

4.1 Targets That Reflect Operating Conditions

Many labels are computed under simplified assumptions. Future datasets should encode potential, pH, electrolyte, surface coverage, reconstruction, and uncertainty so that models predict properties closer to operating electrocatalysts.

4.2 Selectivity and Competing Pathways

Energy conversion reactions often involve many products and side reactions. Models should learn pathway competition and selectivity, especially for CO₂RR, NRR, and ORR, rather than treating one adsorption energy as a complete objective.

4.3 Stability and Dynamic Surfaces

Catalyst surfaces can dissolve, oxidize, reconstruct, segregate, or change coverage during operation. Machine learning potentials, microkinetic models, and in situ data can help, but only if validation includes dynamic and durability evidence.

4.4 Closed-Loop Experimental Discovery

Closed-loop electrocatalyst discovery should connect prediction, synthesis, electrochemical testing, and model updates. The loop must record failed experiments and protocol details because negative and noisy measurements are crucial for learning realistic search boundaries.

4.5 Interpretability and Mechanistic Trust

Catalysis researchers need more than ranked candidate lists. Useful models should expose active-site motifs, descriptor relationships, uncertainty, and mechanistic hypotheses that can be tested by computation or experiment.

5 Conclusion

Machine learning for electrocatalyst discovery is moving from descriptor regression toward integrated discovery systems. High-throughput DFT, graph learning, active learning, reaction-specific modeling, and autonomous experiments can accelerate the search for catalysts for water splitting, fuel cells, CO₂ reduction, nitrogen reduction, and other energy-conversion reactions. The key challenge is integration: models must connect surface chemistry, reaction networks, operating conditions, stability, synthesis, uncertainty, and validation. Progress will depend on datasets and workflows that reward experimentally meaningful discovery rather than only retrospective prediction accuracy.

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